

Simplifying Expressions with Square Roots (Part II)

Multiplying Square Roots

① $\sqrt{2} \cdot \sqrt{2}$ ② $\sqrt{6} \cdot \sqrt{6}$ ③ $\sqrt{13} \cdot \sqrt{13}$ ④ $\sqrt{3} \cdot \sqrt{3}$

$$\sqrt{2 \cdot 2}$$

$$\sqrt{6 \cdot 6}$$

$$\sqrt{13 \cdot 13}$$

$$\sqrt{3 \cdot 3}$$

$$\sqrt{4}$$

$$\sqrt{36}$$

$$\sqrt{169}$$

$$\sqrt{9}$$

$$\boxed{2}$$

$$\boxed{6}$$

$$\boxed{13}$$

$$\boxed{3}$$

⑤ $\sqrt{5} \cdot \sqrt{5}$ ⑥ $\sqrt{7} \cdot \sqrt{7}$ ⑦ $\sqrt{8} \cdot \sqrt{8}$ ⑧ $\sqrt{11} \cdot \sqrt{11}$

$$\sqrt{5 \cdot 5}$$

$$\sqrt{7 \cdot 7}$$

$$\sqrt{8 \cdot 8}$$

$$\sqrt{11 \cdot 11}$$

$$\sqrt{25}$$

$$\sqrt{49}$$

$$\sqrt{64}$$

$$\sqrt{121}$$

$$\boxed{5}$$

$$\boxed{7}$$

$$\boxed{8}$$

$$\boxed{11}$$

⑨ $\sqrt{10} \cdot \sqrt{10}$ ⑩ $\sqrt{14} \cdot \sqrt{14}$ ⑪ $\sqrt{15} \cdot \sqrt{15}$ ⑫ $\sqrt{17} \cdot \sqrt{17}$

$$\sqrt{10 \cdot 10}$$

$$\sqrt{14 \cdot 14}$$

$$\sqrt{15 \cdot 15}$$

$$\sqrt{17 \cdot 17}$$

$$\sqrt{100}$$

$$\sqrt{196}$$

$$\sqrt{225}$$

$$\sqrt{289}$$

$$\boxed{10}$$

$$\boxed{14}$$

$$\boxed{15}$$

$$\boxed{17}$$

⑬ $\sqrt{3} \cdot \sqrt{2}$ ⑭ $\sqrt{10} \cdot (-\sqrt{7})$ ⑮ $\sqrt{7} \cdot \sqrt{2}$ ⑯ $-\sqrt{5} \cdot \sqrt{2}$

$$\sqrt{3 \cdot 2}$$

$$-\sqrt{10 \cdot 7}$$

$$\sqrt{7 \cdot 2}$$

$$-\sqrt{5 \cdot 2}$$

$$\boxed{\sqrt{6}}$$

$$\boxed{-\sqrt{70}}$$

$$\boxed{\sqrt{14}}$$

$$\boxed{-\sqrt{10}}$$

⑰ $-\sqrt{11} \cdot (-\sqrt{3})$ ⑱ $\sqrt{3} \cdot \sqrt{6}$ ⑲ $\sqrt{5} \cdot \sqrt{10}$ ⑳ $\sqrt{6} \cdot \sqrt{2}$

$$\sqrt{11 \cdot 3}$$

$$\sqrt{3 \cdot 6}$$

$$\sqrt{5 \cdot 10}$$

$$\sqrt{6 \cdot 2}$$

$$\boxed{\sqrt{33}}$$

$$\sqrt{18}$$

$$\sqrt{50}$$

$$\sqrt{12}$$

$$\sqrt{9 \cdot 2}$$

$$\sqrt{25 \cdot 2}$$

$$\sqrt{4 \cdot 3}$$

$$\sqrt{9} \sqrt{2}$$

$$\sqrt{25} \sqrt{2}$$

$$\sqrt{4} \sqrt{3}$$

$$\boxed{3\sqrt{2}}$$

$$\boxed{5\sqrt{2}}$$

$$\boxed{2\sqrt{3}}$$

㉑ $-\sqrt{8} \cdot \sqrt{5}$ ㉒ $-\sqrt{7} \cdot \sqrt{12}$ ㉓ $(7\sqrt{2})(10)$ ㉔ $(-8)(6\sqrt{10})$

$$-\sqrt{8 \cdot 5}$$

$$-\sqrt{7 \cdot 12}$$

$$\boxed{70\sqrt{2}}$$

$$\boxed{-48\sqrt{10}}$$

$$-\sqrt{40}$$

$$-\sqrt{84}$$

$$-\sqrt{4 \cdot 10}$$

$$-\sqrt{7 \cdot 4 \cdot 3}$$

$$-\sqrt{4} \sqrt{10}$$

$$-\sqrt{4 \cdot 21}$$

$$\boxed{-2\sqrt{10}}$$

$$\boxed{-2\sqrt{21}}$$

㉕ $(-2\sqrt{3})(-5)$ ㉖ $(-9)(6\sqrt{7})$ ㉗ $3\sqrt{5}(-2\sqrt{6})$ ㉘ $(2\sqrt{7})(6\sqrt{5})$

$$\boxed{10\sqrt{3}}$$

$$\boxed{-54\sqrt{7}}$$

$$-6\sqrt{5 \cdot 6}$$

$$12\sqrt{7 \cdot 5}$$

$$\boxed{-6\sqrt{30}}$$

$$\boxed{12\sqrt{35}}$$

$$(29) -2\sqrt{2}(-5\sqrt{3}) \quad (30) 3\sqrt{7}(-4\sqrt{6}) \quad (31) (-2\sqrt{11})(-\sqrt{3}) \quad (32) (-\sqrt{2})(6\sqrt{13})$$

$$10\sqrt{2 \cdot 3} \quad -12\sqrt{7 \cdot 6} \quad 2\sqrt{11 \cdot 3} \quad -6\sqrt{2 \cdot 13}$$

$$\boxed{10\sqrt{6}} \quad \boxed{-12\sqrt{42}} \quad \boxed{2\sqrt{33}} \quad \boxed{-6\sqrt{26}}$$

$$(33) (-3\sqrt{2})(6\sqrt{10}) \quad (34) -2\sqrt{15}(-8\sqrt{3}) \quad (35) (2\sqrt{30})(7\sqrt{5}) \quad (36) 2\sqrt{6}(-8\sqrt{3})$$

$$-18\sqrt{2 \cdot 10} \quad 16\sqrt{15 \cdot 3} \quad 14\sqrt{30 \cdot 5} \quad -16\sqrt{6 \cdot 3}$$

$$-18\sqrt{20} \quad 16\sqrt{45} \quad 14\sqrt{150} \quad -16\sqrt{18}$$

$$-18\sqrt{4 \cdot 5} \quad 16\sqrt{9 \cdot 5} \quad 14\sqrt{25 \cdot 6} \quad -16\sqrt{9 \cdot 2}$$

$$-18\sqrt{4 \cdot 5} \quad 16\sqrt{9 \cdot 5} \quad 14\sqrt{25 \cdot 6} \quad -16\sqrt{9 \cdot 2}$$

$$-18 \cdot 2 \cdot \sqrt{5} \quad 16 \cdot 3 \cdot \sqrt{5} \quad 14 \cdot 5 \cdot \sqrt{6} \quad -16 \cdot 3 \cdot \sqrt{2}$$

$$\boxed{-36\sqrt{5}} \quad \boxed{48\sqrt{5}} \quad \boxed{70\sqrt{6}} \quad \boxed{-48\sqrt{2}}$$

$$(37) -2\sqrt{24}(-5\sqrt{3}) \quad (38) -4\sqrt{12}(-7\sqrt{3}) \quad (39) \sqrt{3}(5+6\sqrt{6}) \quad (40) 4\sqrt{2}(3-6\sqrt{3})$$

$$10\sqrt{3 \cdot 24} \quad 28\sqrt{36} \quad 5\sqrt{3} + 6\sqrt{3 \cdot 6} \quad 12\sqrt{2} - 24\sqrt{3 \cdot 2}$$

$$10\sqrt{72} \quad 28 \cdot 6 \quad 5\sqrt{3} + 6\sqrt{18} \quad \boxed{12\sqrt{2} - 24\sqrt{6}}$$

$$10\sqrt{36 \cdot 2} \quad \boxed{168} \quad 5\sqrt{3} + 6\sqrt{9 \cdot 2}$$

$$10\sqrt{36 \cdot 2} \quad 5\sqrt{3} + 6\sqrt{9 \cdot 2}$$

$$10 \cdot 6 \cdot \sqrt{2} \quad 5\sqrt{3} + 6 \cdot 3 \cdot \sqrt{2}$$

$$\boxed{60\sqrt{2}} \quad \boxed{5\sqrt{3} + 18\sqrt{2}}$$

$$(41) 2\sqrt{2}(3\sqrt{6}-7\sqrt{3}-5\sqrt{7})$$

$$(42) 5\sqrt{2}(3\sqrt{6}+7\sqrt{2}-10\sqrt{7})$$

$$6\sqrt{2 \cdot 6} - 14\sqrt{2 \cdot 3} - 10\sqrt{2 \cdot 7}$$

$$15\sqrt{2 \cdot 6} + 35\sqrt{2 \cdot 2} - 50\sqrt{2 \cdot 7}$$

$$6\sqrt{12} - 14\sqrt{6} - 10\sqrt{14}$$

$$15\sqrt{12} + 35\sqrt{4} - 50\sqrt{14}$$

$$6\sqrt{4 \cdot 3} - 14\sqrt{6} - 10\sqrt{14}$$

$$15\sqrt{4 \cdot 3} + 35 \cdot 2 - 50\sqrt{14}$$

$$6\sqrt{4 \cdot 3} - 14\sqrt{6} - 10\sqrt{14}$$

$$15\sqrt{4 \cdot 3} + 70 - 50\sqrt{14}$$

$$6 \cdot 2 \cdot \sqrt{3} - 14\sqrt{6} - 10\sqrt{14}$$

$$15 \cdot 2 \cdot \sqrt{3} + 70 - 50\sqrt{14}$$

$$\boxed{12\sqrt{3} - 14\sqrt{6} - 10\sqrt{14}}$$

$$\boxed{30\sqrt{3} + 70 - 50\sqrt{14}}$$

Dividing Square Roots

$$(43) \frac{\sqrt{15}}{\sqrt{3}}$$

$$(44) \frac{\sqrt{20}}{-\sqrt{5}}$$

$$(45) \frac{-\sqrt{56}}{-\sqrt{7}}$$

$$(46) \frac{\sqrt{12}}{\sqrt{2}}$$

$$\sqrt{\frac{15}{3}}$$

$$-\sqrt{\frac{20}{5}}$$

$$\sqrt{\frac{56}{7}}$$

$$\sqrt{\frac{12}{2}}$$

$$\boxed{\sqrt{5}}$$

$$-\sqrt{4}$$

$$\sqrt{8}$$

$$\boxed{\sqrt{6}}$$

$$\boxed{-2}$$

$$\sqrt{4 \cdot 2}$$

$$\sqrt{4 \cdot 2}$$

$$\boxed{2\sqrt{2}}$$

$$\textcircled{47} \frac{-\sqrt{80}}{\sqrt{5}} \quad \textcircled{48} \frac{\sqrt{60}}{\sqrt{5}} \quad \textcircled{49} \frac{-\sqrt{75}}{-\sqrt{15}} \quad \textcircled{50} \frac{\sqrt{75}}{-\sqrt{3}}$$

$$-\sqrt{\frac{80}{5}} \quad \sqrt{\frac{60}{5}} \quad \sqrt{\frac{75}{15}} \quad -\sqrt{\frac{75}{3}}$$

$$-\sqrt{16} \quad \sqrt{12} \quad \boxed{\sqrt{5}} \quad -\sqrt{25}$$

$$\boxed{-4} \quad \sqrt{4 \cdot 3} \quad \boxed{-5}$$

$$\sqrt{4 \cdot \sqrt{3}}$$

$$\boxed{2\sqrt{3}}$$

Rationalizing Denominators

$$\textcircled{51} \frac{3}{\sqrt{5}} \quad \textcircled{52} \frac{2}{\sqrt{3}} \quad \textcircled{53} \frac{4}{\sqrt{5}} \quad \textcircled{54} \frac{8}{\sqrt{6}} \quad \textcircled{55} \frac{10}{\sqrt{7}} \quad \textcircled{56} \frac{6}{\sqrt{2}}$$

$$\frac{3 \cdot \sqrt{5}}{\sqrt{5} \cdot \sqrt{5}} \quad \frac{2 \cdot \sqrt{3}}{\sqrt{3} \cdot \sqrt{3}} \quad \frac{4 \cdot \sqrt{5}}{\sqrt{5} \cdot \sqrt{5}} \quad \frac{8 \cdot \sqrt{6}}{\sqrt{6} \cdot \sqrt{6}} \quad \frac{10 \cdot \sqrt{7}}{\sqrt{7} \cdot \sqrt{7}} \quad \frac{6 \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{2}}$$

$$\frac{3\sqrt{5}}{\sqrt{5 \cdot 5}} \quad \frac{2\sqrt{3}}{\sqrt{3 \cdot 3}} \quad \frac{4\sqrt{5}}{\sqrt{5 \cdot 5}} \quad \frac{8\sqrt{6}}{\sqrt{6 \cdot 6}} \quad \frac{10\sqrt{7}}{\sqrt{7 \cdot 7}} \quad \frac{6\sqrt{2}}{\sqrt{2 \cdot 2}}$$

$$\frac{3\sqrt{5}}{\sqrt{25}} \quad \frac{2\sqrt{3}}{\sqrt{9}} \quad \frac{4\sqrt{5}}{\sqrt{25}} \quad \frac{8\sqrt{6}}{\sqrt{36}} \quad \frac{10\sqrt{7}}{\sqrt{49}} \quad \frac{6\sqrt{2}}{\sqrt{4}}$$

$$\boxed{\frac{3\sqrt{5}}{5}}$$

$$\boxed{\frac{2\sqrt{3}}{3}}$$

$$\boxed{\frac{4\sqrt{5}}{5}}$$

$$\frac{8\sqrt{6}}{6}$$

$$\boxed{\frac{10\sqrt{7}}{7}}$$

$$\frac{6\sqrt{2}}{2}$$

$$\boxed{\frac{4\sqrt{6}}{3}}$$

$$\boxed{3\sqrt{2}}$$

$$\textcircled{57} \frac{3}{\sqrt{24}} \quad \textcircled{58} \frac{-6}{\sqrt{45}} \quad \textcircled{59} \frac{2}{\sqrt{54}} \quad \textcircled{60} \frac{9}{\sqrt{125}}$$

$$\frac{3}{\sqrt{4 \cdot 6}} \quad \frac{-6}{\sqrt{9 \cdot 5}} \quad \frac{2}{\sqrt{9 \cdot 6}} \quad \frac{9}{\sqrt{25 \cdot 5}}$$

$$\frac{3}{\sqrt{4 \cdot \sqrt{6}}} \quad \frac{-6}{\sqrt{9 \cdot \sqrt{5}}} \quad \frac{2}{\sqrt{9 \cdot \sqrt{6}}} \quad \frac{9}{\sqrt{25 \cdot \sqrt{5}}}$$

$$\frac{3}{2\sqrt{6}} \quad \frac{-6}{3\sqrt{5}} \quad \frac{2}{3\sqrt{6}} \quad \frac{9}{5\sqrt{5}}$$

$$\frac{3 \cdot \sqrt{6}}{2\sqrt{6} \cdot \sqrt{6}} \quad \frac{-2}{\sqrt{5}} \quad \frac{2 \cdot \sqrt{6}}{3\sqrt{6} \cdot \sqrt{6}} \quad \frac{9 \cdot \sqrt{5}}{5\sqrt{5} \cdot \sqrt{5}}$$

$$\frac{3\sqrt{6}}{2\sqrt{6 \cdot 6}} \quad \frac{-2 \cdot \sqrt{5}}{\sqrt{5} \cdot \sqrt{5}} \quad \frac{2\sqrt{6}}{3\sqrt{6 \cdot 6}} \quad \frac{9\sqrt{5}}{5\sqrt{5 \cdot 5}}$$

$$\frac{3\sqrt{6}}{2\sqrt{36}} \quad \frac{-2\sqrt{5}}{\sqrt{5 \cdot 5}} \quad \frac{2\sqrt{6}}{3\sqrt{36}} \quad \frac{9\sqrt{5}}{5\sqrt{25}}$$

$$\frac{3\sqrt{6}}{2 \cdot 6} \quad \frac{-2\sqrt{5}}{\sqrt{25}} \quad \frac{2\sqrt{6}}{3 \cdot 6} \quad \frac{9\sqrt{5}}{5 \cdot 5}$$

$$\frac{3\sqrt{6}}{12} \quad \boxed{\frac{-2\sqrt{5}}{5}} \quad \frac{2\sqrt{6}}{18} \quad \boxed{\frac{9\sqrt{5}}{25}}$$

$$\boxed{\frac{\sqrt{6}}{4}}$$

$$\boxed{\frac{\sqrt{6}}{9}}$$

① $\frac{6}{\sqrt{28}}$	② $\frac{8}{\sqrt{18}}$	③ $\frac{-1}{\sqrt{63}}$	④ $\frac{10}{\sqrt{75}}$
$\frac{6}{\sqrt{4 \cdot 7}}$	$\frac{8}{\sqrt{9 \cdot 2}}$	$\frac{-1}{\sqrt{9 \cdot 7}}$	$\frac{10}{\sqrt{25 \cdot 3}}$
$\frac{6}{\sqrt{4} \cdot \sqrt{7}}$	$\frac{8}{\sqrt{9} \sqrt{2}}$	$\frac{-1}{\sqrt{9} \sqrt{7}}$	$\frac{10}{\sqrt{25} \cdot \sqrt{3}}$
$\frac{6}{2\sqrt{7}}$	$\frac{8}{3\sqrt{2}}$	$\frac{-1}{3\sqrt{7}}$	$\frac{10}{5\sqrt{3}}$
$\frac{6 \cdot \sqrt{7}}{2\sqrt{7} \cdot \sqrt{7}}$	$\frac{8 \cdot \sqrt{2}}{3\sqrt{2} \cdot \sqrt{2}}$	$\frac{-1 \cdot \sqrt{7}}{3\sqrt{7} \cdot \sqrt{7}}$	$\frac{10 \cdot \sqrt{3}}{5\sqrt{3} \cdot \sqrt{3}}$
$\frac{6\sqrt{7}}{2\sqrt{7 \cdot 7}}$	$\frac{8\sqrt{2}}{3\sqrt{2 \cdot 2}}$	$\frac{-\sqrt{7}}{3\sqrt{7 \cdot 7}}$	$\frac{10\sqrt{3}}{5\sqrt{3 \cdot 3}}$
$\frac{6\sqrt{7}}{2\sqrt{49}}$	$\frac{8\sqrt{2}}{3\sqrt{4}}$	$\frac{-\sqrt{7}}{3\sqrt{49}}$	$\frac{10\sqrt{3}}{5\sqrt{9}}$
$\frac{6\sqrt{7}}{2 \cdot 7}$	$\frac{8\sqrt{2}}{3 \cdot 2}$	$\frac{-\sqrt{7}}{3 \cdot 7}$	$\frac{10\sqrt{3}}{5 \cdot 3}$
$\frac{6\sqrt{7}}{14}$	$\frac{8\sqrt{2}}{6}$	$\frac{-\sqrt{7}}{21}$	$\frac{10\sqrt{3}}{15}$
$\boxed{\frac{3\sqrt{7}}{7}}$	$\boxed{\frac{4\sqrt{2}}{3}}$		$\frac{2\sqrt{3}}{3}$

⑤ $\sqrt{\frac{5}{4}}$	⑥ $\sqrt{\frac{41}{9}}$	⑦ $\sqrt{\frac{6}{25}}$	⑧ $\sqrt{\frac{10}{49}}$
$\frac{\sqrt{5}}{\sqrt{4}}$	$\frac{\sqrt{41}}{\sqrt{9}}$	$\frac{\sqrt{6}}{\sqrt{25}}$	$\frac{\sqrt{10}}{\sqrt{49}}$
$\boxed{\frac{\sqrt{5}}{2}}$	$\boxed{\frac{\sqrt{41}}{3}}$	$\boxed{\frac{\sqrt{6}}{5}}$	$\boxed{\frac{\sqrt{10}}{7}}$
⑨ $\sqrt{\frac{3}{81}}$	⑩ $\sqrt{\frac{19}{36}}$	⑪ $\sqrt{\frac{9}{2}}$	⑫ $\sqrt{\frac{64}{5}}$
$\frac{\sqrt{3}}{\sqrt{81}}$	$\frac{\sqrt{19}}{\sqrt{36}}$	$\frac{\sqrt{9}}{\sqrt{2}}$	$\frac{\sqrt{64}}{\sqrt{5}}$
$\boxed{\frac{\sqrt{3}}{9}}$	$\boxed{\frac{\sqrt{19}}{6}}$	$\frac{3}{\sqrt{2}}$	$\frac{8}{\sqrt{5}}$
		$\frac{3 \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{2}}$	$\frac{8 \cdot \sqrt{5}}{\sqrt{5} \cdot \sqrt{5}}$
		$\frac{3\sqrt{2}}{\sqrt{2 \cdot 2}}$	$\frac{8\sqrt{5}}{\sqrt{5 \cdot 5}}$
		$\frac{3\sqrt{2}}{\sqrt{4}}$	$\frac{8\sqrt{5}}{\sqrt{25}}$
		$\boxed{\frac{3\sqrt{2}}{2}}$	$\boxed{\frac{8\sqrt{5}}{5}}$

$$\textcircled{73} \sqrt{\frac{49}{6}} \quad \textcircled{74} \sqrt{\frac{16}{7}} \quad \textcircled{75} \sqrt{\frac{100}{11}} \quad \textcircled{76} \sqrt{\frac{36}{37}}$$

$$\frac{\sqrt{49}}{\sqrt{6}}$$

$$\frac{\sqrt{16}}{\sqrt{7}}$$

$$\frac{\sqrt{100}}{\sqrt{11}}$$

$$\frac{\sqrt{36}}{\sqrt{37}}$$

$$\frac{7}{\sqrt{6}}$$

$$\frac{4}{\sqrt{7}}$$

$$\frac{10}{\sqrt{11}}$$

$$\frac{6}{\sqrt{37}}$$

$$\frac{7 \cdot \sqrt{6}}{\sqrt{6} \cdot \sqrt{6}}$$

$$\frac{4 \cdot \sqrt{7}}{\sqrt{7} \cdot \sqrt{7}}$$

$$\frac{10 \cdot \sqrt{11}}{\sqrt{11} \cdot \sqrt{11}}$$

$$\frac{6 \cdot \sqrt{37}}{\sqrt{37} \cdot \sqrt{37}}$$

$$\frac{7\sqrt{6}}{\sqrt{36}}$$

$$\frac{4\sqrt{7}}{\sqrt{49}}$$

$$\frac{10\sqrt{11}}{\sqrt{121}}$$

$$\frac{6\sqrt{37}}{37}$$

$$\boxed{\frac{7\sqrt{6}}{6}}$$

$$\boxed{\frac{4\sqrt{7}}{7}}$$

$$\boxed{\frac{10\sqrt{11}}{11}}$$

$$\textcircled{77} \sqrt{\frac{6}{5}} \quad \textcircled{78} \sqrt{\frac{5}{7}} \quad \textcircled{79} \sqrt{\frac{3}{10}} \quad \textcircled{80} \sqrt{\frac{2}{11}}$$

$$\frac{\sqrt{6}}{\sqrt{5}}$$

$$\frac{\sqrt{5}}{\sqrt{7}}$$

$$\frac{\sqrt{3}}{\sqrt{10}}$$

$$\frac{\sqrt{2}}{\sqrt{11}}$$

$$\frac{\sqrt{6} \cdot \sqrt{5}}{\sqrt{5} \cdot \sqrt{5}}$$

$$\frac{\sqrt{5} \cdot \sqrt{7}}{\sqrt{7} \cdot \sqrt{7}}$$

$$\frac{\sqrt{3} \cdot \sqrt{10}}{\sqrt{10} \cdot \sqrt{10}}$$

$$\frac{\sqrt{2} \cdot \sqrt{11}}{\sqrt{11} \cdot \sqrt{11}}$$

$$\frac{\sqrt{6 \cdot 5}}{\sqrt{5 \cdot 5}}$$

$$\frac{\sqrt{5 \cdot 7}}{\sqrt{7 \cdot 7}}$$

$$\frac{\sqrt{3 \cdot 10}}{\sqrt{10 \cdot 10}}$$

$$\frac{\sqrt{2 \cdot 11}}{\sqrt{11 \cdot 11}}$$

$$\frac{\sqrt{30}}{\sqrt{25}}$$

$$\frac{\sqrt{35}}{\sqrt{49}}$$

$$\frac{\sqrt{30}}{\sqrt{100}}$$

$$\frac{\sqrt{22}}{\sqrt{121}}$$

$$\boxed{\frac{\sqrt{30}}{5}}$$

$$\boxed{\frac{\sqrt{35}}{7}}$$

$$\boxed{\frac{\sqrt{30}}{10}}$$

$$\boxed{\frac{\sqrt{22}}{11}}$$

$$\textcircled{81} \sqrt{\frac{2}{3}} \quad \textcircled{82} \sqrt{\frac{10}{13}} \quad \textcircled{83} \sqrt{\frac{12}{5}} \quad \textcircled{84} \sqrt{\frac{40}{7}}$$

$$\frac{\sqrt{2}}{\sqrt{3}}$$

$$\frac{\sqrt{10}}{\sqrt{13}}$$

$$\frac{\sqrt{12}}{\sqrt{5}}$$

$$\frac{\sqrt{40}}{\sqrt{7}}$$

$$\frac{\sqrt{2} \cdot \sqrt{3}}{\sqrt{3} \cdot \sqrt{3}}$$

$$\frac{\sqrt{10} \cdot \sqrt{13}}{\sqrt{13} \cdot \sqrt{13}}$$

$$\frac{\sqrt{4 \cdot 3}}{\sqrt{5}}$$

$$\frac{\sqrt{4 \cdot 10}}{\sqrt{7}}$$

$$\frac{\sqrt{2 \cdot 3}}{\sqrt{3 \cdot 3}}$$

$$\frac{\sqrt{10 \cdot 13}}{\sqrt{13 \cdot 13}}$$

$$\frac{\sqrt{4} \sqrt{3}}{\sqrt{5}}$$

$$\frac{\sqrt{4} \sqrt{10}}{\sqrt{7}}$$

$$\frac{\sqrt{6}}{\sqrt{9}}$$

$$\frac{\sqrt{130}}{\sqrt{169}}$$

$$\frac{2\sqrt{3}}{\sqrt{5}}$$

$$\frac{2\sqrt{10}}{\sqrt{7}}$$

$$\boxed{\frac{\sqrt{6}}{3}}$$

$$\boxed{\frac{\sqrt{130}}{13}}$$

$$\frac{2\sqrt{3} \cdot \sqrt{5}}{\sqrt{5} \cdot \sqrt{5}}$$

$$\frac{2\sqrt{10} \cdot \sqrt{7}}{\sqrt{7} \cdot \sqrt{7}}$$

$$\frac{2\sqrt{3 \cdot 5}}{\sqrt{5 \cdot 5}}$$

$$\frac{2\sqrt{10 \cdot 7}}{\sqrt{7 \cdot 7}}$$

$$\frac{2\sqrt{15}}{\sqrt{25}}$$

$$\frac{2\sqrt{70}}{\sqrt{49}}$$

$$\boxed{\frac{2\sqrt{15}}{5}}$$

$$\boxed{\frac{2\sqrt{70}}{7}}$$

$$\textcircled{85} \sqrt{\frac{8}{3}} \quad \textcircled{86} \sqrt{\frac{20}{11}} \quad \textcircled{87} \sqrt{\frac{48}{5}} \quad \textcircled{88} \sqrt{\frac{24}{7}}$$

$$\frac{\sqrt{8}}{\sqrt{3}}$$

$$\frac{\sqrt{20}}{\sqrt{11}}$$

$$\frac{\sqrt{48}}{\sqrt{5}}$$

$$\frac{\sqrt{4 \cdot 2}}{\sqrt{3}}$$

$$\frac{\sqrt{4 \cdot 5}}{\sqrt{11}}$$

$$\frac{\sqrt{16 \cdot 3}}{\sqrt{5}}$$

$$\frac{\sqrt{4} \sqrt{2}}{\sqrt{3}}$$

$$\frac{\sqrt{4} \cdot \sqrt{5}}{\sqrt{11}}$$

$$\frac{\sqrt{16} \cdot \sqrt{3}}{\sqrt{5}}$$

$$\frac{2\sqrt{2}}{\sqrt{3}}$$

$$\frac{2\sqrt{5}}{\sqrt{11}}$$

$$\frac{4\sqrt{3}}{\sqrt{5}}$$

$$\frac{2\sqrt{2} \cdot \sqrt{3}}{\sqrt{3} \cdot \sqrt{3}}$$

$$\frac{2\sqrt{5} \cdot \sqrt{11}}{\sqrt{11} \cdot \sqrt{11}}$$

$$\frac{4\sqrt{3} \cdot \sqrt{5}}{\sqrt{5} \cdot \sqrt{5}}$$

$$\frac{2\sqrt{2 \cdot 3}}{\sqrt{3 \cdot 3}}$$

$$\frac{2\sqrt{5 \cdot 11}}{\sqrt{11 \cdot 11}}$$

$$\frac{4\sqrt{3 \cdot 5}}{\sqrt{5 \cdot 5}}$$

$$\frac{2\sqrt{6}}{\sqrt{9}}$$

$$\frac{2\sqrt{55}}{\sqrt{121}}$$

$$\frac{4\sqrt{15}}{\sqrt{25}}$$

$$\boxed{\frac{2\sqrt{6}}{3}}$$

$$\boxed{\frac{2\sqrt{55}}{11}}$$

$$\boxed{\frac{4\sqrt{15}}{5}}$$

$$\textcircled{89} \sqrt{\frac{7}{18}} \quad \textcircled{90} \sqrt{\frac{5}{12}} \quad \textcircled{91} \sqrt{\frac{2}{45}} \quad \textcircled{92} \sqrt{\frac{2}{27}}$$

$$\frac{\sqrt{7}}{\sqrt{18}}$$

$$\frac{\sqrt{5}}{\sqrt{12}}$$

$$\frac{\sqrt{2}}{\sqrt{45}}$$

$$\frac{\sqrt{7}}{\sqrt{9 \cdot 2}}$$

$$\frac{\sqrt{5}}{\sqrt{4 \cdot 3}}$$

$$\frac{\sqrt{2}}{\sqrt{9 \cdot 5}}$$

$$\frac{\sqrt{7}}{\sqrt{9 \cdot 2}}$$

$$\frac{\sqrt{5}}{\sqrt{4 \cdot 3}}$$

$$\frac{\sqrt{2}}{\sqrt{9 \cdot 5}}$$

$$\frac{\sqrt{7}}{3\sqrt{2}}$$

$$\frac{\sqrt{5}}{2\sqrt{3}}$$

$$\frac{\sqrt{2}}{3\sqrt{5}}$$

$$\frac{\sqrt{7} \cdot \sqrt{2}}{3\sqrt{2} \cdot \sqrt{2}}$$

$$\frac{\sqrt{5} \cdot \sqrt{3}}{2\sqrt{3} \cdot \sqrt{3}}$$

$$\frac{\sqrt{2} \cdot \sqrt{5}}{3\sqrt{5} \cdot \sqrt{5}}$$

$$\frac{\sqrt{7 \cdot 2}}{3\sqrt{2 \cdot 2}}$$

$$\frac{\sqrt{5 \cdot 3}}{2\sqrt{3 \cdot 3}}$$

$$\frac{\sqrt{2 \cdot 5}}{3\sqrt{5 \cdot 5}}$$

$$\frac{\sqrt{14}}{3\sqrt{4}}$$

$$\frac{\sqrt{15}}{2\sqrt{9}}$$

$$\frac{\sqrt{10}}{3\sqrt{25}}$$

$$\frac{\sqrt{14}}{3 \cdot 2}$$

$$\frac{\sqrt{15}}{2 \cdot 3}$$

$$\frac{\sqrt{10}}{3 \cdot 5}$$

$$\boxed{\frac{\sqrt{14}}{6}}$$

$$\boxed{\frac{\sqrt{15}}{6}}$$

$$\boxed{\frac{\sqrt{10}}{15}}$$

$$\textcircled{93} \sqrt{\frac{10}{63}} \quad \textcircled{94} \sqrt{\frac{7}{99}} \cdot \textcircled{95} \sqrt{\frac{27}{20}} \quad \textcircled{96} \sqrt{\frac{44}{45}}$$

$$\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$$

$$\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

$$1^2, 2^2, 3^2, 4^2, 5^2, 6^2, 7^2, 8^2, 9^2, 10^2$$

$$1, 4, 9, 16, 25, 36, 49, 64, 81, 100$$

$$\frac{\sqrt{27}}{\sqrt{20}}$$

$$\frac{\sqrt{44}}{\sqrt{45}}$$

$$\frac{\sqrt{9 \cdot 3}}{\sqrt{4 \cdot 5}}$$

$$\frac{\sqrt{4 \cdot 11}}{\sqrt{9 \cdot 5}}$$

$$\frac{\sqrt{9 \cdot 3}}{\sqrt{4 \cdot 5}}$$

$$\frac{\sqrt{4 \cdot 11}}{\sqrt{9 \cdot 5}}$$

$$\frac{3\sqrt{3}}{2\sqrt{5}}$$

$$\frac{2\sqrt{11}}{3\sqrt{5}}$$

$$\frac{3\sqrt{3 \cdot 5}}{2\sqrt{5 \cdot 5}}$$

$$\frac{2\sqrt{11 \cdot 5}}{3\sqrt{5 \cdot 5}}$$

$$\frac{3\sqrt{3 \cdot 5}}{2\sqrt{5 \cdot 5}}$$

$$\frac{2\sqrt{11 \cdot 5}}{3\sqrt{5 \cdot 5}}$$

$$\frac{3\sqrt{15}}{2\sqrt{25}}$$

$$\frac{2\sqrt{55}}{3\sqrt{25}}$$

$$\frac{3\sqrt{15}}{2 \cdot 5}$$

$$\frac{2\sqrt{55}}{3 \cdot 5}$$

$$\boxed{\frac{3\sqrt{15}}{10}}$$

$$\boxed{\frac{2\sqrt{55}}{15}}$$

$$\textcircled{97} \sqrt{\frac{63}{56}} \quad \textcircled{98} \sqrt{\frac{50}{27}} \quad \textcircled{99} \sqrt{\frac{27}{28}} \quad \textcircled{100} \sqrt{\frac{48}{75}}$$

$$\frac{\sqrt{63}}{\sqrt{56}}$$

$$\frac{\sqrt{9 \cdot 7}}{\sqrt{4 \cdot 14}}$$

$$\frac{\sqrt{9 \cdot 7}}{\sqrt{4 \cdot 14}}$$

$$\frac{3\sqrt{7}}{2\sqrt{14}}$$

$$\frac{3\sqrt{7 \cdot 14}}{2\sqrt{14 \cdot 14}}$$

$$\frac{3\sqrt{7 \cdot 14}}{2\sqrt{14 \cdot 14}}$$

$$\frac{3\sqrt{98}}{2\sqrt{196}}$$

$$\frac{3\sqrt{98}}{2 \cdot 14}$$

$$\frac{3\sqrt{98}}{28}$$

$$\frac{3\sqrt{49 \cdot 2}}{28}$$

$$\frac{3\sqrt{49 \cdot 2}}{28}$$

$$\frac{3 \cdot 7 \cdot \sqrt{2}}{28}$$

$$\frac{21\sqrt{2}}{28}$$

$$\boxed{\frac{3\sqrt{2}}{4}}$$

$$\begin{array}{r} 7.14 \\ 14 \overline{) 98.2} \\ \underline{98} \\ 0 \end{array}$$

$$49.2$$